# The Blessings of Overparameterization: Applications in Solving Economic Models

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# Motivation, Question, and Contribution

# Motivation

**"I remember my friend Johnny von Neumann used to say, with four parameters I can fit an elephant, and with five I can make him wiggle his trunk."**—*Enrico Fermi*

- Deep learning has proved remarkably capable at solving **high-dimensional problems**.
- Economists are now using it to find accurate approximate solutions to dynamic models once thought intractable.
- These neural networks are **massively overparameterized**, with far more parameters than "data" points.
- Standard statistical wisdom says: more parameters than "data" points leads to **overfitting** and poor out-of-grid / out-of-distribution performance.

# Question

When solving dynamic models with neural networks:

- Does overparameterization lead to poor approximation in out-of-grid / out-of-distribution performance?
- Is overparameterization a "necessary evil" that comes with modern machine learning methods, or does it give us something extra?

# Contribution

1. **No overfitting:** as the number of network parameters grows, the solutions become more accurate.

2. **Algorithmic stability:** the approximate solutions of dynamic models in economics are unstable; overparameterization increases stability.

**Implication**

When in doubt, use a bigger network. Overparameterization is not a source of risk to be managed but a feature to be exploited.

# Background: Economic Models, Approximate Solutions, and Overparameterization

# An economic model is a system of functional equations

Many theoretical models can be written as functional equations:

- Economic object of interest: $f$, where $f : \mathcal{X} \to \mathcal{R} \subseteq \mathbb{R}^N$
    - e.g., entry/exit decision, investment choice, best-response, etc.
- Domain of $f$: $\mathcal{X}$
    - e.g., market conditions, space of capital, opponents' state, etc.
- The "Economics model": $\ell\big(x, f\big)$
    - e.g., Euler and Bellman residuals, equilibrium FOCs.

Then a **solution** is $f^* \in \mathcal{F}$ where $\ell(x, f^*) = \mathbf{0}$ for all $x \in \mathcal{X}$.

# Approximate solution

1. Sample $\mathcal{X}$: $\mathcal{D} = \{x_1, \cdots, x_N\}$
2. Pick a family of parametric functions (e.g., polynomials, Chebyshev polynomials, neural networks) $f(\cdot;\theta)$:
   - $\theta = \{\theta_1, \cdots, \theta_M\}$: parameters for optimization (e.g., coefficients, or weights and biases).
3. To find an approximation for $f$ solve:

$$\min_{\theta} \frac{1}{N} \sum_{x \in \mathcal{D}} \| \underbrace{\ell\left(x, f(\cdot;\theta)\right)}_{\text{Econ model error}} \|_2^2$$

# Neural Networks and Overparameterization

**Neural Networks** form a **highly-overparameterized** ($M \gg N$) class of functions.

- Example: one layer neural network, $f(\cdot;\theta) : \mathbb{R}^Q \to \mathbb{R}$:

$$f(x;\theta) = W_2 \cdot \sigma\left(W_1 \cdot x + b_1\right) + b_2$$

- $W_1 \in \mathbb{R}^{P \times Q}$, $b_1 \in \mathbb{R}^{P \times 1}$, $W_2 \in \mathbb{R}^{1 \times P}$, and $b_2 \in \mathbb{R}$.
- $\theta \equiv \{b_1, W_1, b_2, W_2\}$ are the coefficients, in this example $M = PQ + P + P + 1$.
- $\sigma(\cdot)$ is a nonlinear function applied element-wise (e.g., $\max\{\cdot, 0\}$).
- Making it "deeper" by adding another "layer": $f(x;\theta) \equiv W_3 \cdot \sigma(W_2 \cdot \sigma(W_1 \cdot x + b_1) + b_2) + b_3$.

# Deep learning is a non-convex optimization

- Training a neural network is **non-convex** in $\theta$ on its own, since the layers compose nonlinear activations.
- On top of that, most economic dynamic models are **nonlinear**.
- In a non-convex problem, the solution depends on the **initialization** of the weights and biases $\theta$.
- So the "answer" can depend on the random seed.
- We run the algorithm for $50$ different initializations (seeds) of the weights and biases.

# Applications

## Experimental design

- We demonstrate these properties across three canonical models with known benchmarks:

**1. Linear-Quadratic**

Closed-form solution. Sharpest test of approximation error.

**2. McCall Job Search**

1D state, kink at reservation wage. Non-smooth approximation.

**3. Real Business Cycle**

2D stochastic state. No closed form. Benchmarked vs. VFI.

- In all three: **accuracy** and **stability** improve with overparameterization.

# Application 1: Linear-Quadratic regulator

# The LQ problem

- The general deterministic LQ problem:

$$v(x) = \max_u \left\{-x^\top R\, x - u^\top Q\, u + \beta\, v(x')\right\}$$
$$\text{s.t.} \quad x' = Ax + Bu, \quad x_0 \text{ given}$$

- State variable: $x \in \mathbb{R}^n$. Control variable: $u \in \mathbb{R}^m$.
- Closed-form solution:

$$\boxed{v(x) = -x^\top P x, \qquad u(x) = -Fx}$$

where $P$ and $F$ are obtained from the discrete-time algebraic Riccati equation.

- Sharpest test: approximation error can be computed exactly at every point.

# A competitive firm with adjustment costs

- A price-taking firm in a competitive industry.
- Own output $y$, aggregate output $Y$ (exogenous). Inverse demand: $\alpha_0 - \alpha_1 Y$.
- The firm chooses investment $u$ subject to quadratic adjustment cost $\frac{\gamma}{2}u^2$:

$$v(y,Y) = \max_u \left\{ (\alpha_0 - \alpha_1 Y)y - \tfrac{\gamma}{2}u^2 + \beta\, v(y', Y') \right\}$$
$$\text{s.t.} \quad Y' = h_0 + h_1 Y, \quad y' = y + u$$

- State is two-dimensional: $(y, Y)$. With augmented state $x = (1, y, Y)^\top$, this maps into the general LQ framework.
- Euler equation:

$$\boxed{\gamma\, u(y,Y) = \beta \left[ \gamma\, u(y', Y') + (\alpha_0 - \alpha_1 Y') \right]}$$

# LQ: neural network implementation

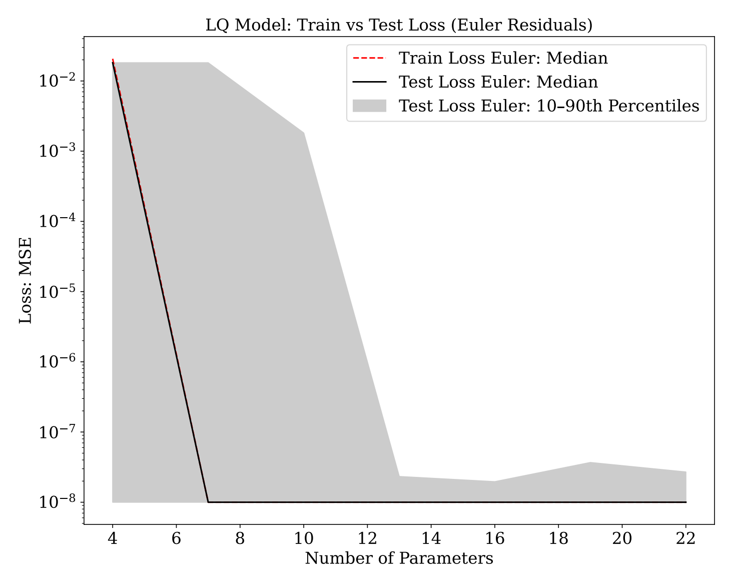
- **Architecture:** one-hidden-layer ReLU networks for value and policy functions. Hidden units $H = 1$ to $H = 7$ (4 to 22 parameters).
- **Training data:** the optimal $u$ depends only on $Y$ ($y$ enters linearly and does not affect the marginal return to investment), so we minimize the mean squared Euler residual over $n_Y = 6$ grid points:

$$\boxed{\min_{\theta_u} \frac{1}{n_Y} \sum_{i=1}^{n_Y} \left[\gamma\, u(Y_i;\theta_u) - \beta\big(\gamma\, u(Y_i';\theta_u) + \alpha_0 - \alpha_1 Y_i'\big)\right]^2}$$

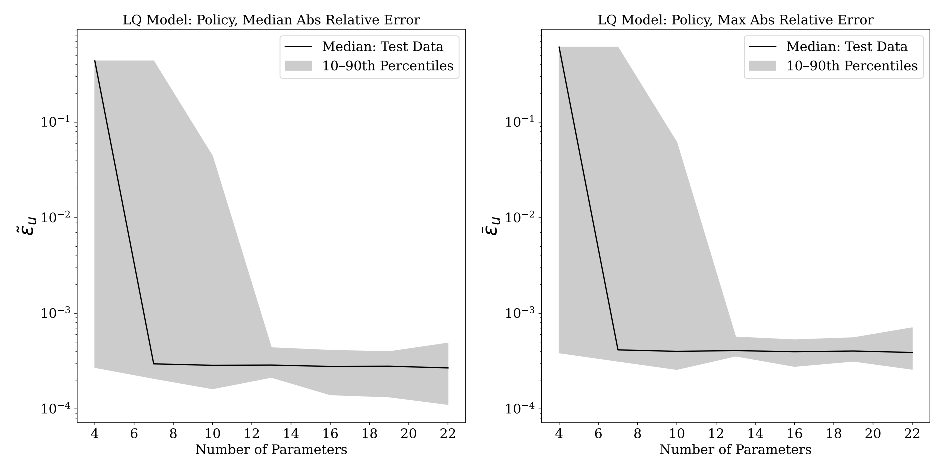
- Adam optimizer with early stopping at loss $< 10^{-8}$.
- **Test data:** $T = 29$ period path simulated from fixed initial conditions. Test points lie in the interior of the state space and never on a training node.

Parameterization

# LQ: train and test Euler-residual loss

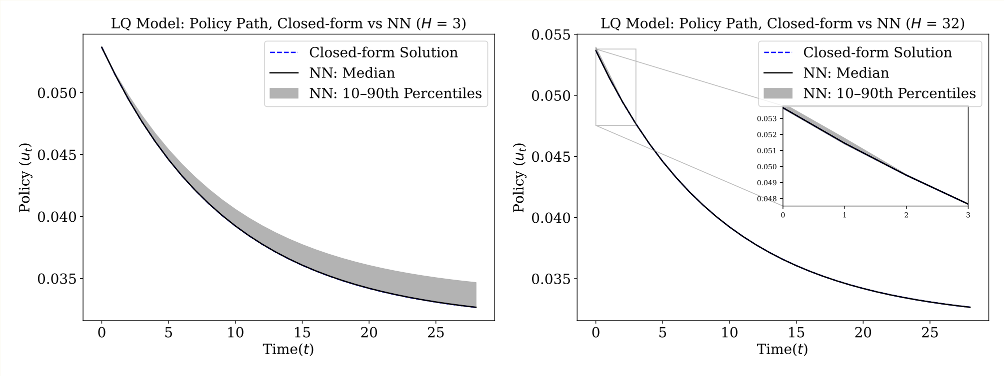


# LQ: absolute relative error vs. closed-form solution




- Median (left) and maximum (right) absolute relative error of the NN policy. Both statistics and their cross-seed dispersion fall with network width.

# LQ: policy paths, underparameterized vs. overparameterized




- Left: $H = 3$ (underparameterized). Median is roughly accurate, but the across-seed band is wide.
- Right: $H = 32$ (overparameterized). Median tracks the closed-form solution closely and the band collapses.

# Application 2: McCall job-search model

# From a linear to a non-smooth operator equation

- The LQ model is linear, which makes it the cleanest test but also the easiest.
- Does the blessing survive when the operator $\mathcal{T}$ is nonlinear and the solution is non-smooth?
- The McCall model (McCall, 1970) provides exactly this test: the value function has a **kink** at the reservation wage, so the network must approximate a non-differentiable function.
- An unemployed worker receives i.i.d. wage offers $w \sim f$ on $[0, B]$ each period.
- Accept: permanent income stream $w/(1-\beta)$. Reject: unemployment compensation $c$ plus continuation value.
- Bellman equation (a nonlinear integral equation with a max operator):

$$\boxed{v(w) = \max\left\{\frac{w}{1-\beta},\ c + \beta \int_0^B v(w')f(w')\,dw'\right\}}$$

## McCall: the kink and why it matters

- Optimal policy: threshold form. There exists a reservation wage $\bar{w}$ such that accept iff $w > \bar{w}$.
- Closed-form value function:

$$v(w) = \begin{cases} \bar{w}/(1-\beta) & \text{if } w \le \bar{w}, \\ w/(1-\beta) & \text{if } w > \bar{w}. \end{cases}$$

- The reservation wage $\bar{w}$ solves:

$$\bar{w} - c = \frac{\beta}{1-\beta} \int_{\bar{w}}^{B} (w' - \bar{w}) f(w')\, dw'$$

  LHS: net gain from rejecting the marginal offer. RHS: expected gain from continued search.
- The kink at $\bar{w}$ is a demanding test: the network must learn a non-smooth function from only $N = 12$ training points.

## McCall: neural network implementation

- **Architecture:** one-hidden-layer ReLU network $v(w;\theta_v)$, linear output. 50 random seeds per width.
- **Training:** $N = 12$ uniform grid on $[0,1]$. Minimize mean squared Bellman residual:

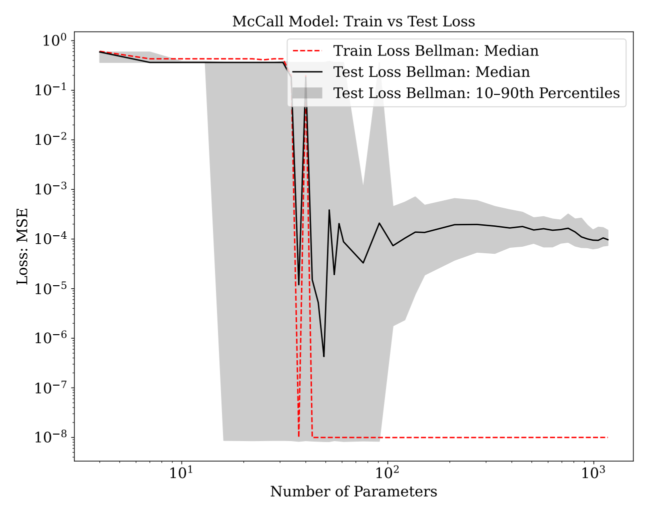
$$\boxed{\min_{\theta_v} \frac{1}{N} \sum_{w_i \in \mathcal{D}} \left[ v(w_i;\theta_v) - \max\left\{ \frac{w_i}{1-\beta},\ c + \beta \int_0^B v(w';\theta_v) f(w')\, dw' \right\} \right]^2}$$

  Integral by quadrature. Adam with early stopping at $10^{-8}$.

- **Test:** finer $N = 100$ uniform grid, strictly outside the training set.
- **Accuracy metric:** absolute relative error $\varepsilon_v(w) = |v(w;\theta_v^*) - v(w)|/|v(w)|$. Summarized by max and median across test grid, then 10th/50th/90th percentiles across 50 seeds.

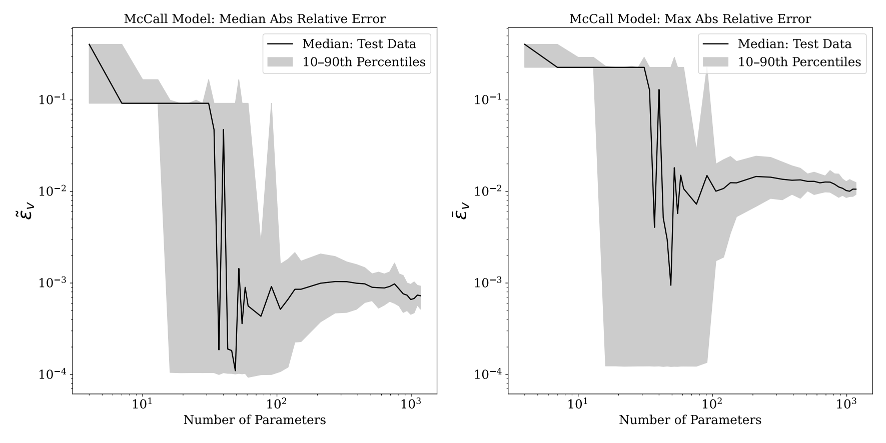
Parameterization

# McCall: train and test Bellman-residual loss



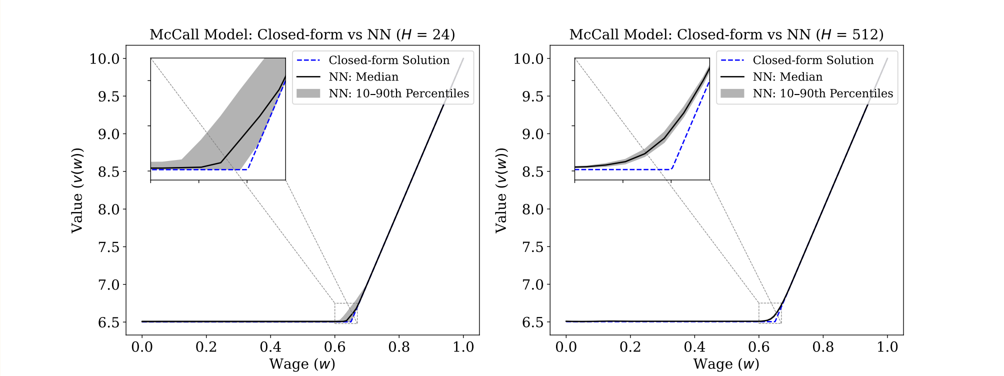

- Test loss descends again in the overparameterized regime: the **double-descent pattern**.
- The across-seed band collapses.

# McCall: absolute relative error




- Median (left) and max (right) relative error fall from ∼30% to ∼2% as the network enters the overparameterized regime.
- The across-seed band narrows sharply: **stability blessing confirmed**.

# McCall: value function, underparameterized vs. overparameterized




- Left: $H = 24$ (underparameterized). Median tracks the closed-form solution well, but the across-seed band is wide.
- Right: $H = 512$ (overparameterized). Band is nearly invisible: 50 random initializations converge to the same function.

# Application 3: Real Business Cycle model

## From deterministic to stochastic, from closed form to none

- The LQ model is linear with a closed-form solution. The McCall model is nonlinear with a kink but still one-dimensional and still admits a closed form.
- The Real Business Cycle model (Kydland and Prescott, 1982) is the most demanding test: a two-dimensional stochastic state space $(k_t, z_t)$ and no analytical solution.
- Social planner, Cobb–Douglas production $e^{z_t} k_t^{\alpha}$, stochastic TFP:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} = e^{z_t} k_t^{\alpha} + (1-\delta) k_t$$

$$z_t = \rho z_{t-1} + \sigma \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$

- Key object: policy function $k'(k, z)$. Benchmark: value function iteration (VFI) on a fine grid.

# RBC: the Euler equation

- Euler equation:

$$\boxed{u'(c_t) = \beta\,\mathbb{E}_t\left[u'(c_{t+1})\left(\alpha e^{z_{t+1}} k_{t+1}^{\alpha-1} + 1 - \delta\right)\right]}$$

- **LHS**: marginal utility cost of saving one unit today.
- **RHS**: expected discounted marginal utility gain from extra capital tomorrow (marginal product plus undepreciated capital).
- Nonlinearity plus stochastic two-dimensional state means no analytical solution.

# RBC: neural network implementation

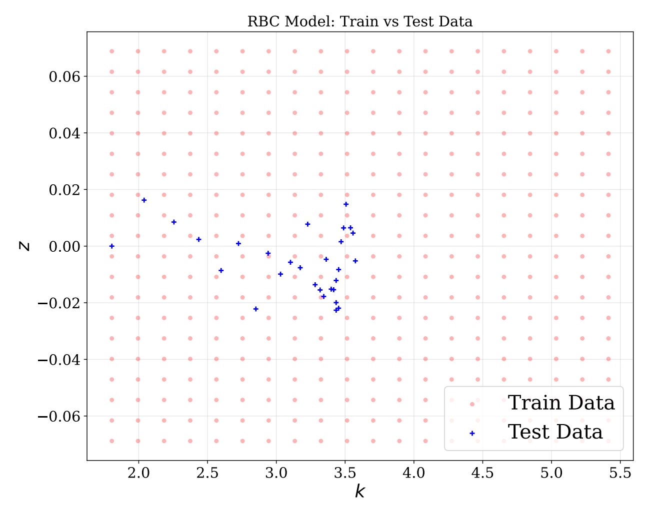
- **Architecture:** one-hidden-layer ReLU network $k'(k, z; \theta)$. Output layer: Softplus activation $\log(1 + e^x)$ to enforce $k' > 0$. 50 seeds per width.
- **Training:** $20 \times 20$ Cartesian grid ($N = 400$). Euler residual:

$$\boxed{\ell_{\mathrm{Euler}}(k, z; \theta) = \frac{1}{c(k, z; \theta)} - \beta \sum_{i=1}^{J} \frac{w_i}{\sqrt{\pi}} \left[ \frac{\alpha e^{\rho z + \sqrt{2}\sigma\zeta_i} k'(k, z; \theta)^{\alpha-1} + 1 - \delta}{c\big(k'(k, z; \theta),\ \rho z + \sqrt{2}\sigma\zeta_i;\ \theta\big)} \right]}$$

  with $J = 15$ Gauss–Hermite nodes. Adam with StepLR scheduler.
- **Penalty term:** $\lambda\big(k'(k^*, 0; \theta) - k^*\big)^2$ with $\lambda = 0.01$, enforcing a necessary implication of the transversality condition at the steady state.

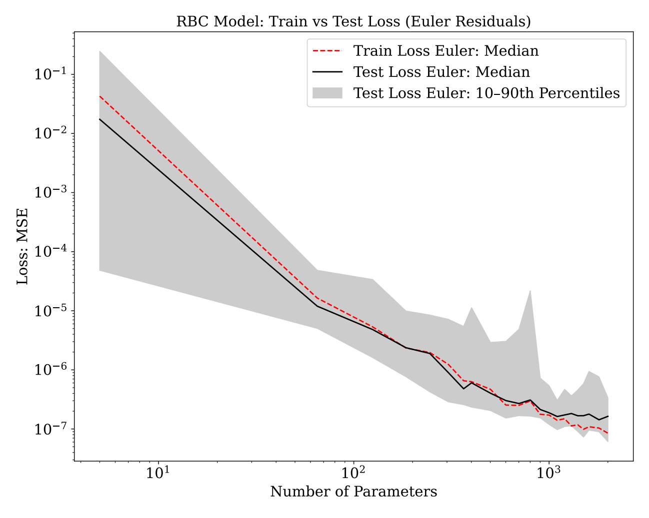
▶▶ Parameterization

# RBC: training and test data



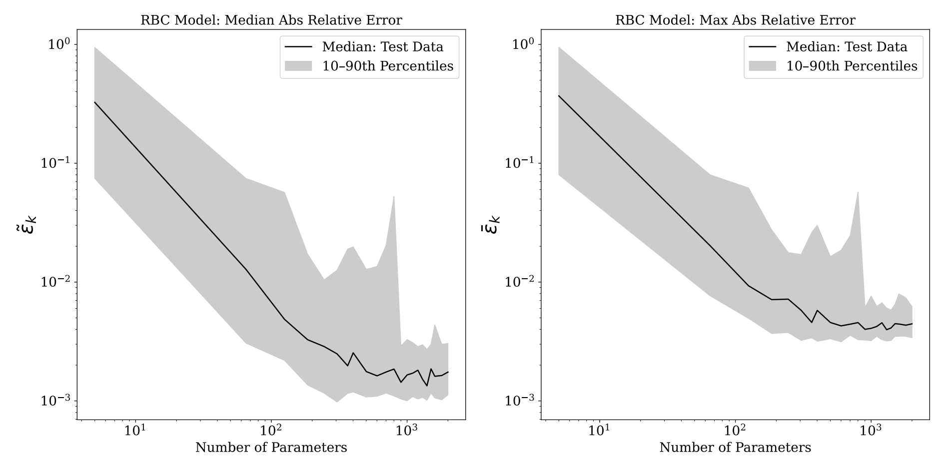

- Red dots: $20 \times 20$ training grid.
- Blue crosses: $T = 29$ path simulated under the VFI policy starting at $k_0 = 0.5\,k^*$.
- Test path visits the interior of the state space and is never on a training node.

# RBC: train and test Euler-residual loss



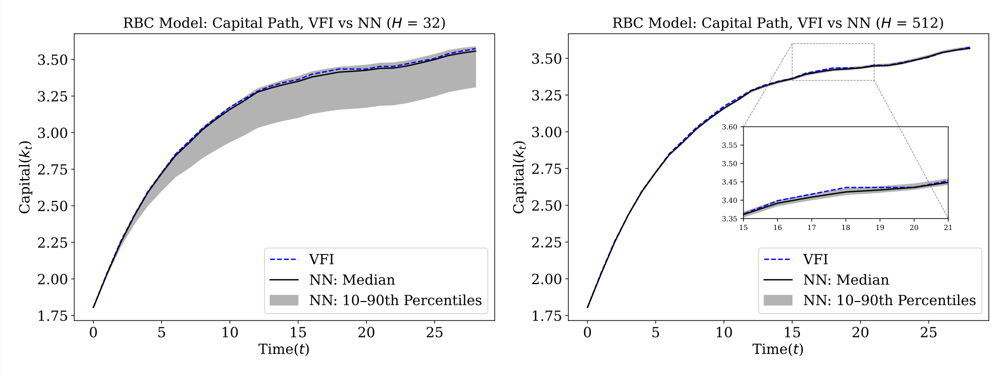

- Both losses decline as the network grows wider, with no **sign of overfitting**.
- The across-seed band narrows sharply: overparameterization simultaneously improves accuracy and reduces initialization sensitivity.

# RBC: absolute relative error vs. VFI




- Median (left) and maximum (right) absolute relative error of the NN capital policy versus VFI.
- Both statistics decline and concentrate around the median seed as width increases.

# RBC: capital paths, underparameterized vs. overparameterized




- Left: $H = 32$. Median path close to VFI, but across-seed band is wide. Accurate on average but unreliable.
- Right: $H = 512$. Median tracks VFI almost exactly over the full horizon and the band collapses to near zero.

# Robustness

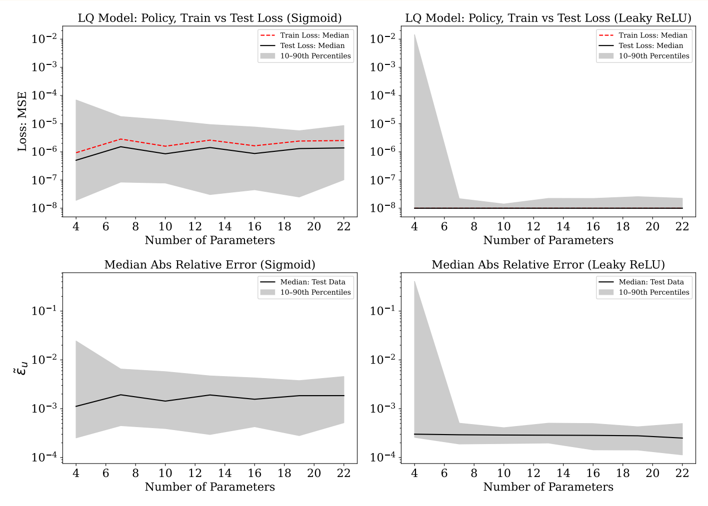
# Are these results specific to ReLU and one hidden layer?

- A natural concern: the blessing might depend on the particular activation function or the single-layer architecture.
- We test robustness along two dimensions:
  - **Activation functions:** Sigmoid and Leaky ReLU in addition to the baseline ReLU.
  - **Network depth:** two-hidden-layer architectures in addition to the baseline one-hidden-layer.

**Result**

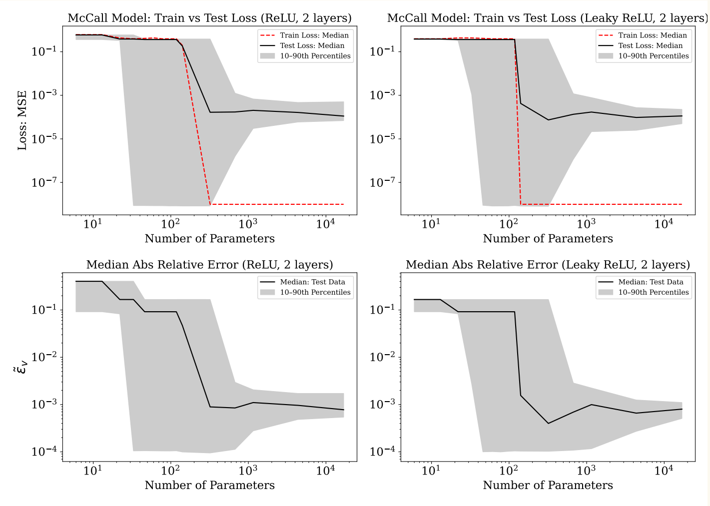
The blessing holds in every configuration we tried. The next slides document this for each model.

# Robustness: alternative activations for the LQ model



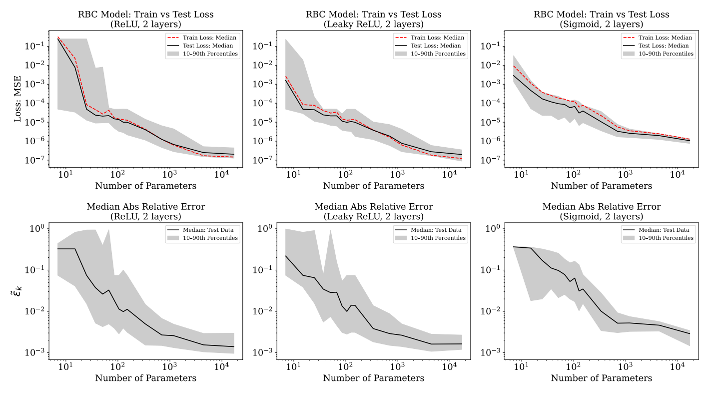

- Sigmoid (left) and Leaky ReLU (right).
- Top row: train/test Euler MSE. Bottom row: median relative error.
- Both activations replicate the main finding: accuracy improves with width, across-seed dispersion collapses.

# Robustness: deeper networks for the McCall model




- Two-hidden-layer architecture with ReLU (left) and Leaky ReLU (right).
- Quantitatively similar results: the blessing is not specific to one-hidden-layer networks.

# Robustness: deeper networks for the RBC model




- Two-hidden-layer architecture with ReLU (left), Leaky ReLU (center), Sigmoid (right).
- All three activations replicate the main finding for a stochastic model with no closed-form solution.

# Robustness: summary

- The blessing of overparameterization is robust across all dimensions most relevant to applied practice:
    - **Activation functions:** ReLU, Leaky ReLU, Sigmoid.
    - **Network depth:** one and two hidden layers.
    - **Operator type:** Euler residual (LQ, RBC), Bellman residual (McCall).
    - **Solution regularity:** smooth (LQ), non-smooth with kink (McCall), stochastic without closed form (RBC).
    - **State dimensionality:** 1D (McCall), 2D (LQ, RBC).

# Conclusion

# Takeaways

- **Finding 1:** no overfitting. As network width increases, out-of-sample Euler and Bellman residuals decline. Value and policy functions converge toward their benchmarks.
- **Finding 2:** algorithmic stability. Small networks produce solutions that depend on the **initialization**. Wide networks concentrate across seeds, making the algorithm reliable and **initialization-independent**.
- Documented consistently across three operator equations: linear with closed form (LQ), nonlinear with kink (McCall), stochastic with no analytical solution (RBC).
- Robust across activations (ReLU, Leaky ReLU, Sigmoid) and depths (one and two hidden layers).

**Bottom line**

When in doubt, use a wider network. Overparameterization is a blessing, not a curse.

# Appendix

# Parameterization: Linear-Quadratic model

| Description | Symbol | Value | |
|---|---|---|---|
| Discount factor | $\beta$ | 0.9 | |
| Demand intercept | $\alpha_0$ | 1.0 | inverse demand level |
| Demand slope | $\alpha_1$ | 1.3 | price sensitivity to aggregate |
| Adjustment cost | $\gamma$ | 100 | convex investment cost |
| Aggregate drift | $h_0$ | 0.05 | constant in $Y' = h_0 + h_1 Y$ |
| Aggregate persistence | $h_1$ | 0.9 | AR coefficient for $Y$ |

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# Parameterization: McCall model

| Description | Symbol | Value | |
|---|---|---|---|
| Discount factor | $\beta$ | 0.9 | |
| Wage support | $B$ | 1.0 | upper bound of $\mathrm{U}[0, B]$ |
| Unemployment benefit | $c$ | 0.1 | flow payoff while searching |

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# Parameterization: Real Business Cycle model

| Description | Symbol | Value | |
|---|---|---|---|
| Discount factor | $\beta$ | 0.96 | |
| Capital share | $\alpha$ | $1/3$ | Cobb–Douglas exponent |
| Depreciation rate | $\delta$ | 0.1 | per-period capital depreciation |
| TFP persistence | $\rho$ | 0.9 | AR(1) coefficient for $z_t$ |
| TFP volatility | $\sigma$ | 0.01 | std. dev. of $\varepsilon_t$ |
| Utility function | $u(c)$ | $\ln c$ | log utility |

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